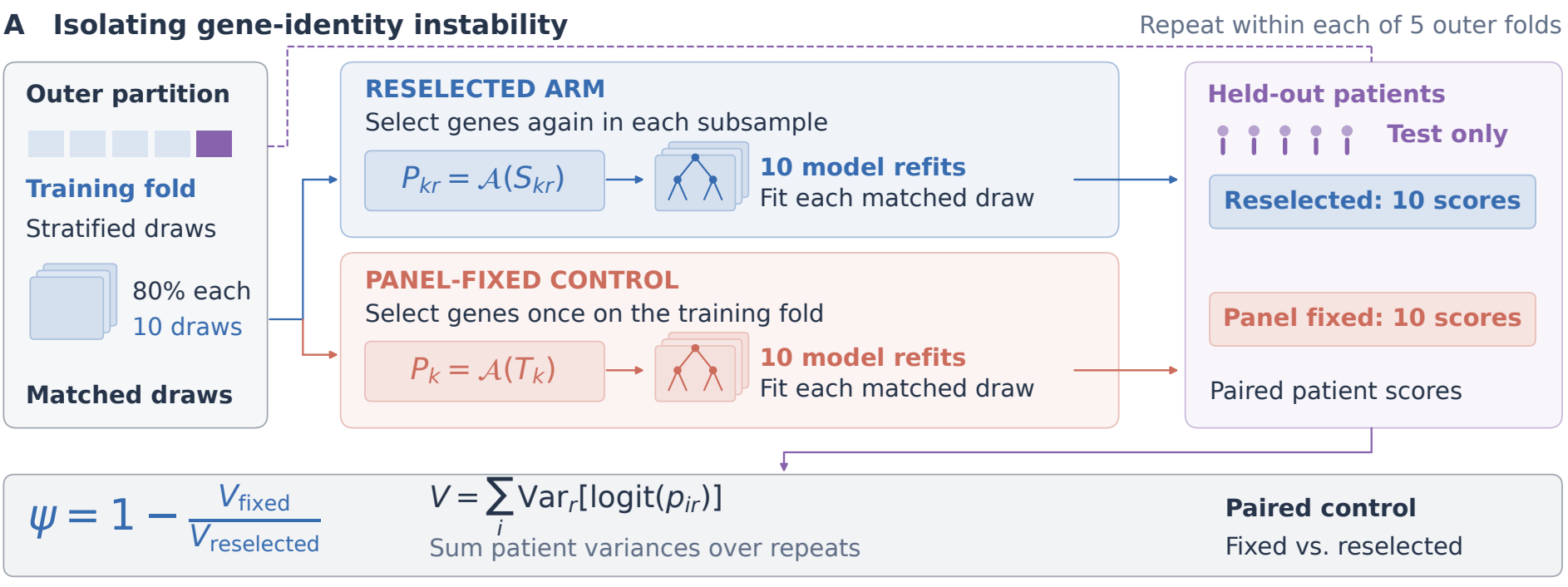
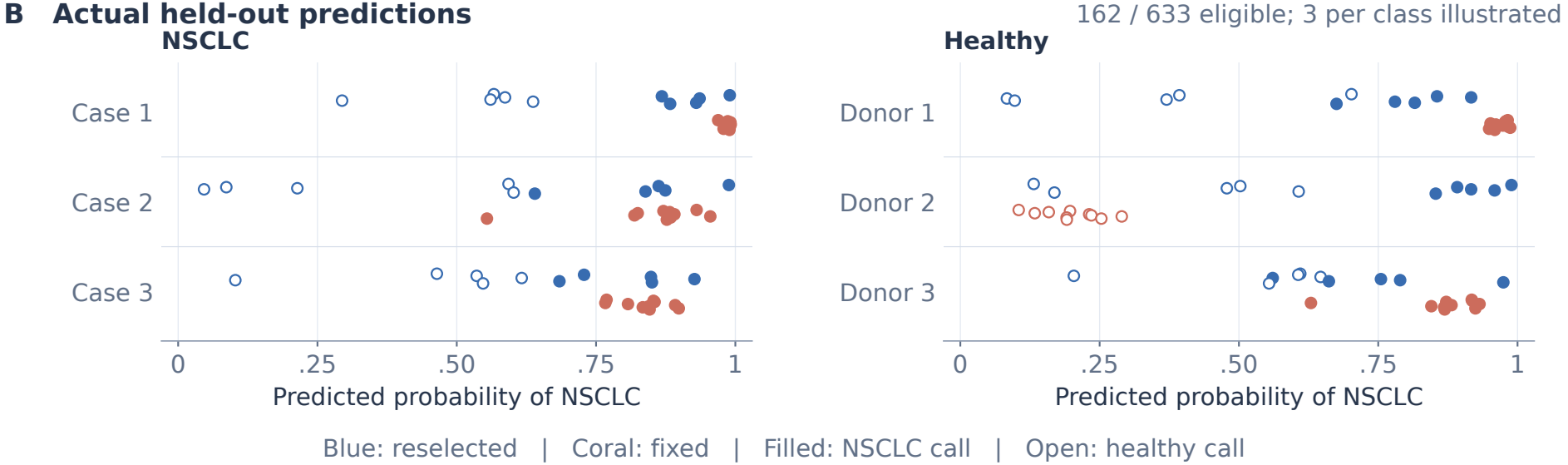

A Isolating gene-identity instability
Repeat within each of 5 outer folds
Outer partition
Training fold
Stratified draws
80% each
10 draws
Matched draws
RESELECTED ARM
Select genes again in each subsample
$P_{kr} = \mathcal{A}(S_{kr})$
10 model refits
Fit each matched draw
PANEL-FIXED CONTROL
Select genes once on the training fold
$P_k = \mathcal{A}(T_k)$
10 model refits
Fit each matched draw
Held-out patients
Test only
Reselected: 10 scores
Panel fixed: 10 scores
Paired patient scores
$\psi = 1 - \frac{V_{\text{fixed}}}{V_{\text{reselected}}}$
$V = \sum_i \text{Var}_r[\text{logit}(p_{ir})]$
Sum patient variances over repeats
Paired control
Fixed vs. reselected


B Actual held-out predictions
162 / 633 eligible; 3 per class illustrated
NSCLC
Case 1
Case 2
Case 3
0
.25
.50
.75
1
Predicted probability of NSCLC
Healthy
Donor 1
Donor 2
Donor 3
0
.25
.50
.75
1
Predicted probability of NSCLC
Blue: reselected | Coral: fixed | Filled: NSCLC call | Open: healthy call